Generative AI (BCA1501) – Unit 1 Full Detailed Answers

This PDF contains exam-ready long answers for Unit-1 questions of BCA1501 Generative AI. The answers are written in simple English with headings, explanations, examples, and diagrams suitable for university examinations.

Q1. Define Artificial Intelligence and differentiate between Narrow AI and General AI.

Artificial Intelligence (AI) is a branch of computer science that develops systems capable of performing tasks that normally require human intelligence such as learning, reasoning, problem solving, understanding language, and decision making.

Narrow AI: Designed for a specific task. Examples: Google Translate, face recognition, spam filters, Siri.
General AI: A hypothetical AI with human-like intelligence that can learn and perform any intellectual task. It does not yet exist.

Difference: Narrow AI has limited scope, while General AI has broad adaptable intelligence.

Q2. Historical evolution of AI from symbolic systems to Transformers.

1950s–1970s: Symbolic AI and rule-based expert systems.
1980s: Expert systems became popular in medicine and engineering.
1990s: Machine learning and statistical methods gained importance.
2000s: Large datasets and computing power enabled deep learning.
2012: Deep neural networks achieved breakthroughs in image recognition.
2017: Transformer architecture introduced self-attention and parallel processing.
2018 onward: BERT, GPT, and other Large Language Models transformed natural language processing.

Q3. Working of a Transformer model with labeled diagram.

Textual diagram:
Input Tokens → Embedding → Positional Encoding → Multi-Head Self-Attention → Feed Forward Network → Output Probabilities.

Self-attention allows each word to focus on relevant words in the sentence. Positional encoding provides word order information. Multi-head attention captures different relationships simultaneously.

Q4. How LLMs differ from rule-based AI systems.

Rule-based AI: Uses manually written IF–THEN rules, has limited flexibility, and performs poorly on unseen situations.
LLMs: Learn patterns from massive text datasets, generate new text, answer questions, summarize documents, translate languages, and adapt to many tasks without explicit rules.

Q5. Role of attention mechanism over RNN/LSTM.

RNNs process tokens sequentially and struggle with long-term dependencies. LSTMs improve memory but still process sequentially. Attention allows the model to directly focus on important words regardless of distance, improves long-context understanding, enables parallel computation, and scales better for large models.

Q6. Training of Large Language Models.

LLMs are trained on enormous text corpora collected from books, websites, articles, and documents. Text is tokenized, the model predicts the next token, errors are measured using a loss function, and parameters are updated through backpropagation and gradient descent. Training is repeated for billions of examples on GPU/TPU clusters.

Advantages: broad knowledge, strong language understanding, text generation, summarization, translation, and adaptability through fine-tuning.

Q7. Tokenization with example.

Tokenization converts text into smaller units called tokens.

Sentence: *Artificial intelligence is powerful.*
Word tokens: [Artificial] [intelligence] [is] [powerful] [.]
Subword tokens may split a word into smaller meaningful parts.

Importance: Converts human language into numerical representations that neural networks can process efficiently.

Unit-1 Quick Revision

- AI = machines performing intelligent tasks.
- Narrow AI is task-specific; General AI is hypothetical human-level AI.
- Transformers use self-attention and positional encoding.
- LLMs learn from data, unlike rule-based systems.
- Attention improves long-sequence modeling and parallelization.
- Tokenization converts text into processable tokens.

End of Unit-1